



AI Powered Tableau Generator

Hasan Ahmed Shaik, Shyam Prasad Mannem, Renuka Mohanta, Wichita State University



Introduction

- **Problem** : Data Analysts spend ~40% of their time on designing dashboard instead of analyzing. Creating dashboard required expertise and time
- **Solution:** Automated Tableau Workbooks (.twb/.twbx) using AI to analyze data an create charts, KPI's and visualization
- **How It Works** : Upload CSV/Excel -> Auto-detect column types -> Assess data quality -> Recommend KPIs and charts (using AI or rule-based recommendation).
- Sponsored by **Industry Advisory Board**; public datasets (Kaggle) used to build AI & dashboard skills.
- We learned to use LangGraph, Pydantic and Streamlit to create a pipeline which generates automated dashboards.

Architecutre &Process

The system uses a Lang Graph orchestration pipeline with six sequential stages:

1. **Validate** — Verify file format, check encoding, and ensure data readability.
2. **Profile** — Detect column types, compute summary s statistics, and generate a quality score.
3. **Analyze** — Use LLM (Google Gemini-3 Flash) to identify key metrics and dimensions. After 3 retries, falls back to rule-based analysis for reliability.
4. **Recommend** — Select optimal chart types (bar, line, pie, scatter) and KPIs based on data characteristics.
5. **Generate** — Produce Tableau XML workbook (.twb/.twbx) with recommended visualizations.
6. **Finalize** — Package outputs, generate logs, and present dashboard for download.

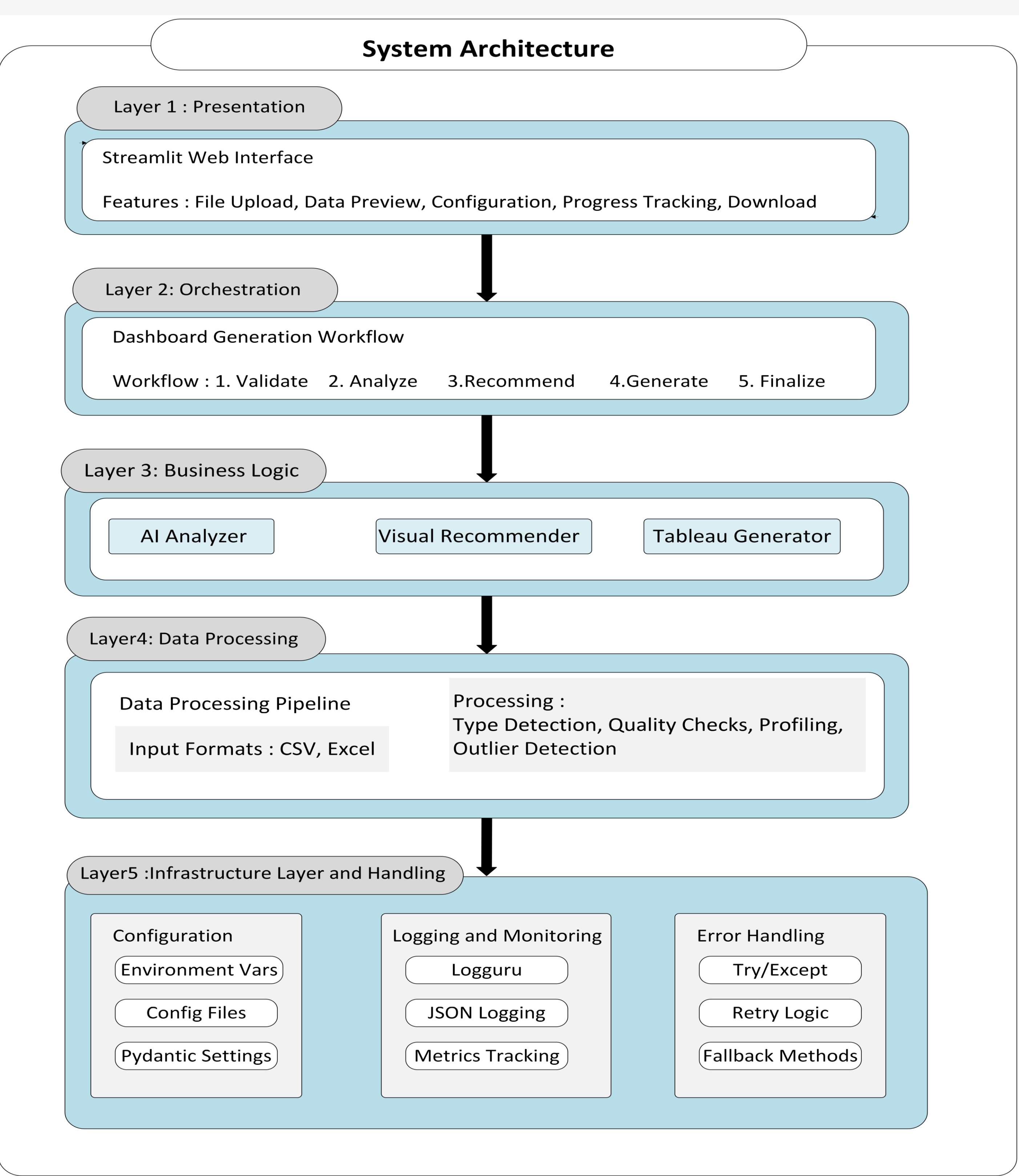
Core Modules

- **DataProcessor:** Handles data validation, type detection, and quality assessment using Pydantic schemas for type safety.
- **DashboardAnalyzer:** Hybrid LLM + rule-based fallback engine for KPI and chart recommendations. Uses JSON schema prompts for consistent outputs.
- **TableauWorkbookGenerator:** Generates valid Tableau XML output with auto-configured worksheets and dashboards using ElementTree.

Reliability & Fallback

- When the LLM fails (network issues, malformed responses), the system automatically retries up to 3
- times, then switches to a deterministic rule-based fallback. This hybrid approach ensures the pipeline never fails silently

Our Architecture will help you understand how each layer worked and it’s responsibilities in the diagram



Results

Dashboard Generation Outputs

The system successfully generates valid .twb/.twbx files containing auto-detected KPIs, recommended chart types, and a data quality score (100-0). Generated workbooks can be directly opened in Tableau Desktop.

Addressing Project Goals

The primary question -"Can AI generate professional Tableau dashboards from raw data?" The pipeline produces complete, importable Tableau workbooks with:

- Automatically selected KPIs based on data profiling
- Appropriate chart types (bar, line, pie) matched to column types
- Quality-scored data with flagged issues for review
- Full traceability via JSON logs at each pipeline stage

Traceability & Logging :

JSON output is generated at each pipeline stage for full traceability. Loguru provides structured logging for every module, enabling debugging and complete audit trails.

```
{} dashboard_specjson ×
C: > Users > shaik > Desktop > Course Material > Semester3 > CS896_Data_Science_Capstone_Project > Project_Bkp > logs > {}
3 "visuals": [
34 {
35   "chart_type": "line",
36   "title": "Monthly Sales Trend",|
37   "x": "Order Date",
38   "y": "Sales",
39   "aggregation": "sum",
40   "group_by": null,
41   "rationale": "Trend metric from AI analysis.",
42   "confidence": 0.75
43 },
44 {
45   "chart_type": "bar",
46   "title": "Sales by Ship Mode",
47   "x": "Ship Mode",
48   "y": "Sales",
49   "aggregation": "sum",
50   "group_by": null,
51   "rationale": "Comparative metric from AI analysis.",
52   "confidence": 0.72
53 },
54 ]
}
```

Visual Recommender JSON Output

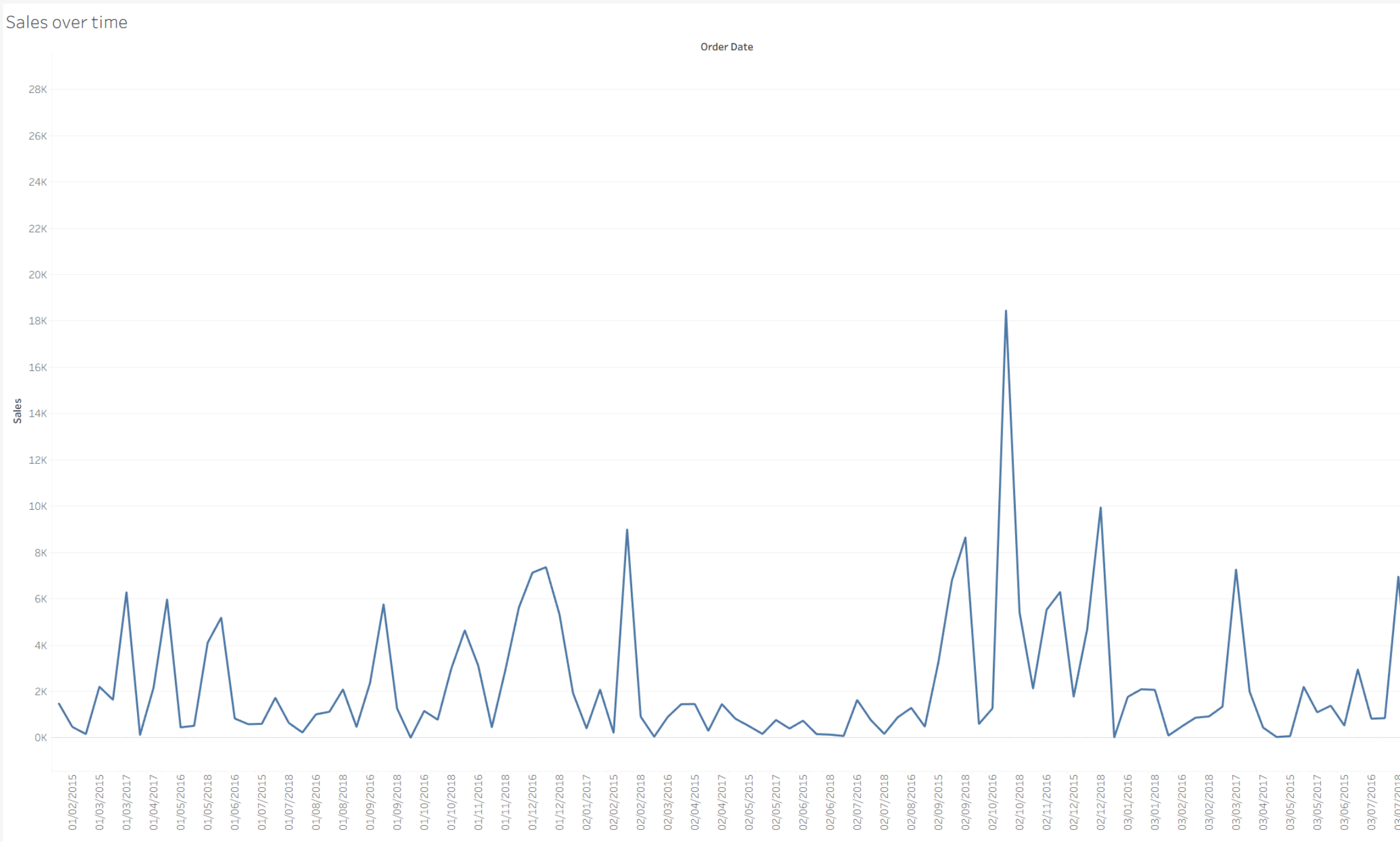
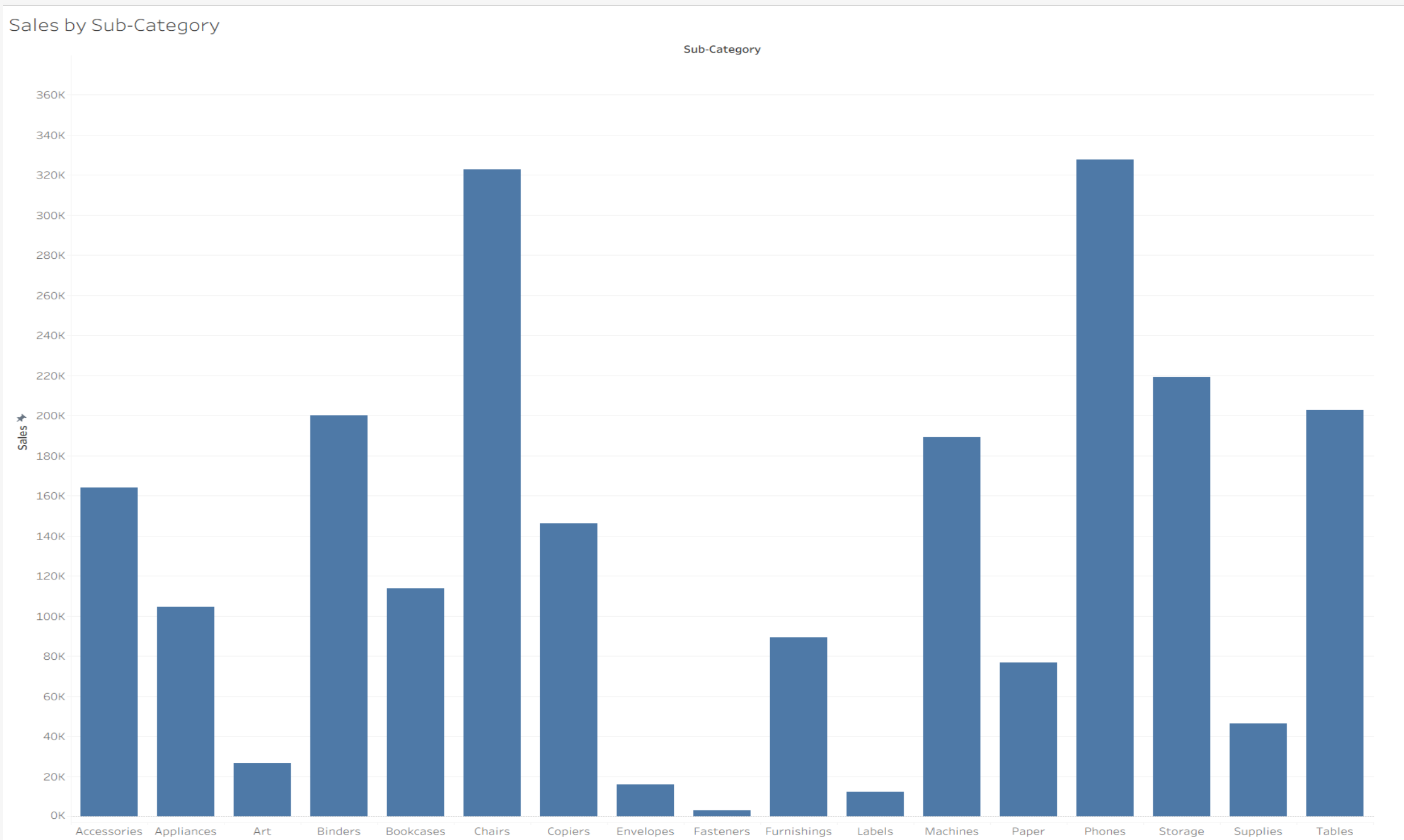


Chart : Sales Over Time



Sales by Subcategory

Discussion

Lessons Learned

- AI dashboard generation is feasible; hybrid LLM + rules ensures reliability when AI fails
- JSON schema prompts, Pydantic constraints, Tableau XML Generation, Fallback mechanism for production grade LLM applications

Future Work:

- Expand chart types (treemaps, etc.) in Tableau XML
- Add more LLMs; test on larger & varied datasets
- NLP for text columns; custom CI/CD pipeline